

CO2 AT2- Simulation Assesment

NAME: A. Prabhakaran

Reg no: 192412577

Course code: ECA5614

1. Simulation of Control Plane and User Plane Separation under Packet Delay, Signaling Load, Throughput, Latency, and Efficiency.

Aim: To simulate the separation of the control plane and user plane in a 5G core network and analyze the effects of packet delay on signaling load, throughput, latency, and overall network efficiency using MATLAB.

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Program :

```
clc;
clear;
close all;

packet_delay = 10:10:100;

signaling_load = 0.5*packet_delay;
throughput = 1000./(1+packet_delay/50);
latency = packet_delay + 5;
efficiency = throughput./(signaling_load+1);

figure;

subplot(2,3,1)
plot(packet_delay,signaling_load,'LineWidth',2)
```

```
xlabel('Packet Delay (ms)')
```

```
ylabel('Signaling Load')
```

```
subplot(2,3,2)
```

```
plot(packet_delay,throughput,'LineWidth',2)
```

```
xlabel('Packet Delay (ms)')
```

```
ylabel('Throughput (Mbps)')
```

```
subplot(2,3,3)
```

```
plot(packet_delay,latency,'LineWidth',2)
```

```
xlabel('Packet Delay (ms)')
```

```
ylabel('Latency (ms)')
```

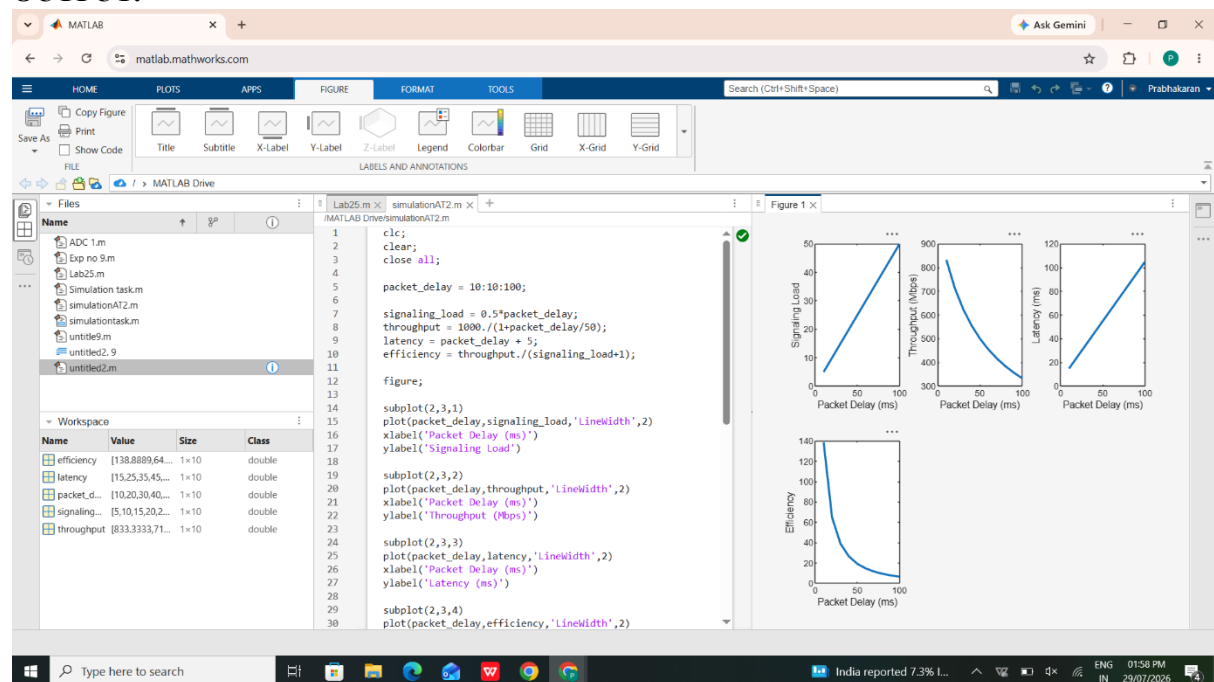
```
subplot(2,3,4)
```

```
plot(packet_delay,efficiency,'LineWidth',2)
```

```
xlabel('Packet Delay (ms)')
```

```
ylabel('Efficiency')
```

OUTPUT:



RESULT:

The simulation was successfully carried out in MATLAB. The results showed that as packet delay increased, signaling load and latency increased, while throughput and network efficiency decreased. This demonstrates the impact of control and user plane separation on 5G network performance.

2. Simulation of 5G Core Traffic Routing under User Density, Mobility, Latency, Throughput, and Resource Utilization.

AIM: To simulate traffic routing in the 5G core network under varying user densities and evaluate its impact on mobility, latency, throughput, and resource utilization using MATLAB.

CODE: `clc;`

`clear;`

`close all;`

`user_density = 100:100:1000;`

`mobility = user_density/100;`

`latency = 5 + user_density/200;`

`throughput = 1200 - user_density*0.4;`

`resource_utilization = user_density/10;`

`figure`

`subplot(2,2,1)`

`plot(user_density,mobility,'LineWidth',2)`

`xlabel('User Density')`

`ylabel('Mobility')`

`subplot(2,2,2)`

`plot(user_density,latency,'LineWidth',2)`

`xlabel('User Density')`

```
ylabel('Latency')
```

```
subplot(2,2,3)
```

```
plot(user_density,throughput,'LineWidth',2)
```

```
xlabel('User Density')
```

```
ylabel('Throughput')
```

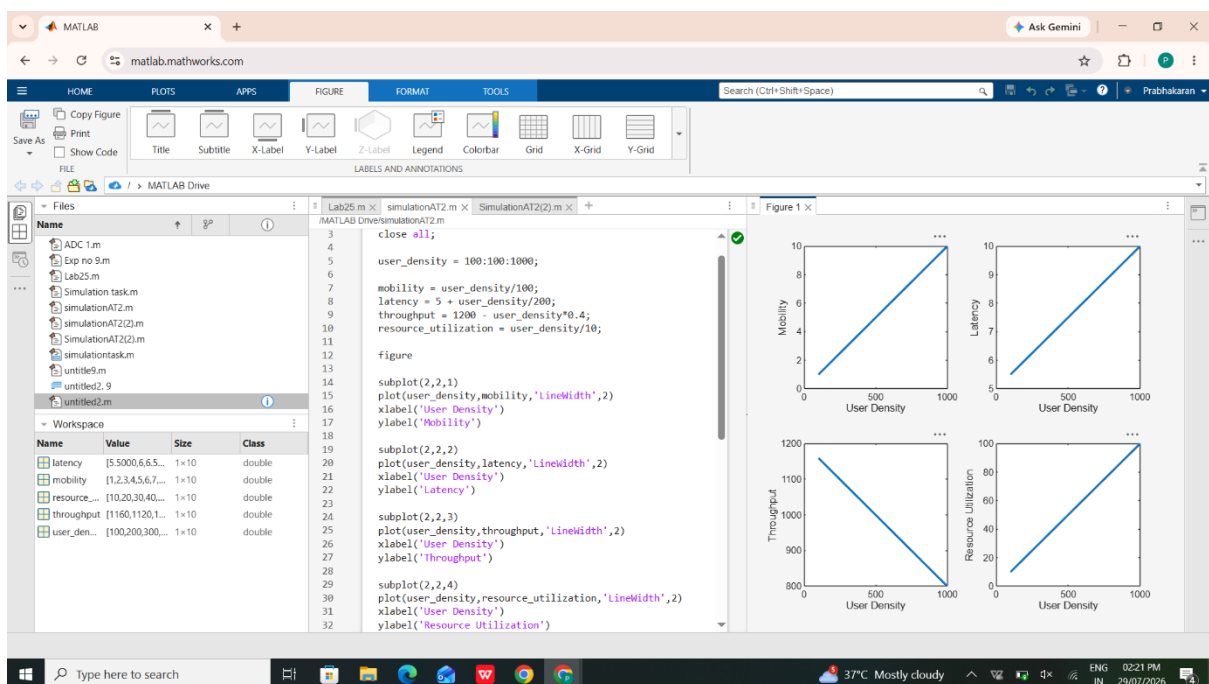
```
subplot(2,2,4)
```

```
plot(user_density,resource_utilization,'LineWidth',2)
```

```
xlabel('User Density')
```

```
ylabel('Resource Utilization')
```

OUTPUT:



RESULT:

The MATLAB simulation was completed successfully. The results indicated that increasing user density led to higher mobility, latency, and resource utilization, whereas throughput gradually decreased due to increased network traffic.

3.Simulation of Packet Delivery Performance in 5G Core under Load Variation, Routing Delay, Throughput, Packet Loss, and Queue Size.

AIM: To simulate packet delivery performance in a 5G core network under different traffic loads and analyze routing delay, throughput, packet loss, and queue size using MATLAB.

Program:

```
clc;
clear;
close all;

load_variation = 10:10:100;

routing_delay = load_variation*0.8;
throughput = 1500-load_variation*8;
packet_loss = load_variation*0.15;
queue_size = load_variation*2;

figure

subplot(2,2,1)
plot(load_variation,routing_delay,'LineWidth',2)
xlabel('Load Variation')
ylabel('Routing Delay')

subplot(2,2,2)
plot(load_variation,throughput,'LineWidth',2)
xlabel('Load Variation')
ylabel('Throughput')

subplot(2,2,3)
```

```
plot(load_variation,packet_loss,'LineWidth',2)
```

```
xlabel('Load Variation')
```

```
ylabel('Packet Loss (%)')
```

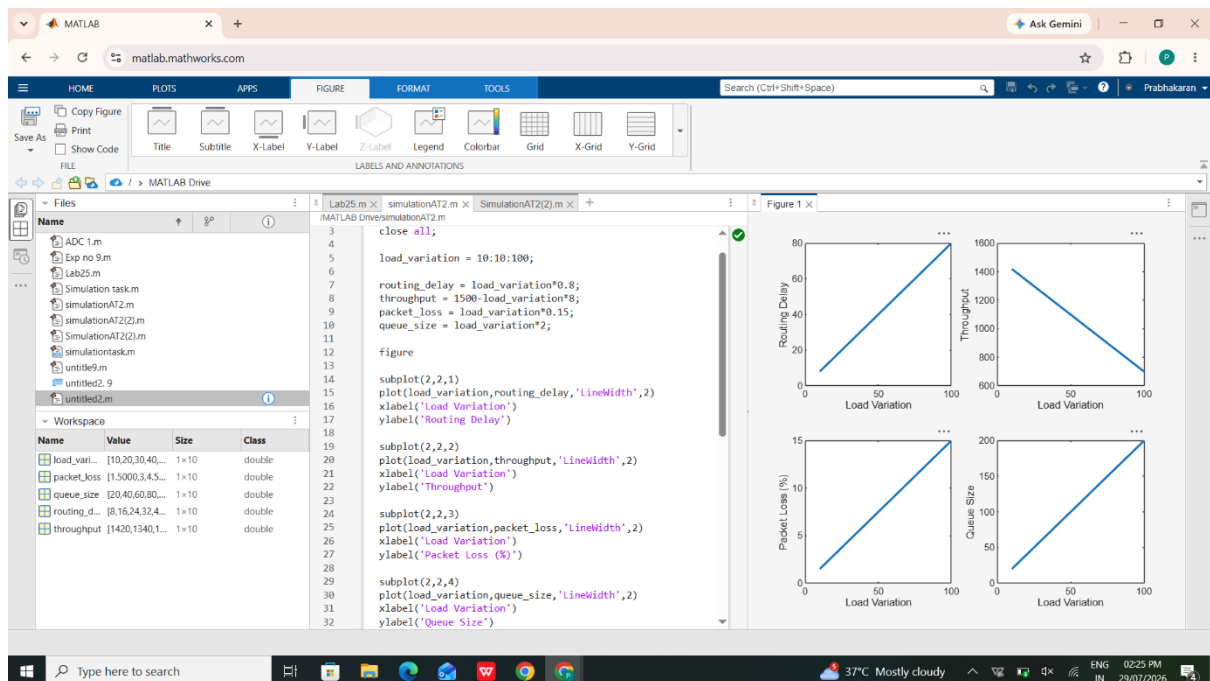
```
subplot(2,2,4)
```

```
plot(load_variation,queue_size,'LineWidth',2)
```

```
xlabel('Load Variation')
```

```
ylabel('Queue Size')
```

OUTPUT:



RESULT:

The simulation was successfully performed in MATLAB. The results showed that increasing traffic load increased routing delay, packet loss, and queue size, while throughput decreased due to network congestion.

4. Simulation of Mobility Management under Handover Count, Delay, Throughput, Signaling Load, and Failure Probability.

AIM: To simulate mobility management in a 5G network by evaluating the effects of handover count on delay, throughput, signaling load, and handover failure probability using MATLAB.

Program:

```
clc;

clear;

close all;

handover_count = 1:20;

delay = handover_count*2;

throughput = 1000-handover_count*20;

signaling_load = handover_count*5;

failure_probability = handover_count/40;

figure

subplot(2,2,1)

plot(handover_count,delay,'LineWidth',2)

xlabel('Handover Count')

ylabel('Delay')

subplot(2,2,2)

plot(handover_count,throughput,'LineWidth',2)

xlabel('Handover Count')

ylabel('Throughput')

subplot(2,2,3)

plot(handover_count,signaling_load,'LineWidth',2)

xlabel('Handover Count')

ylabel('Signaling Load')

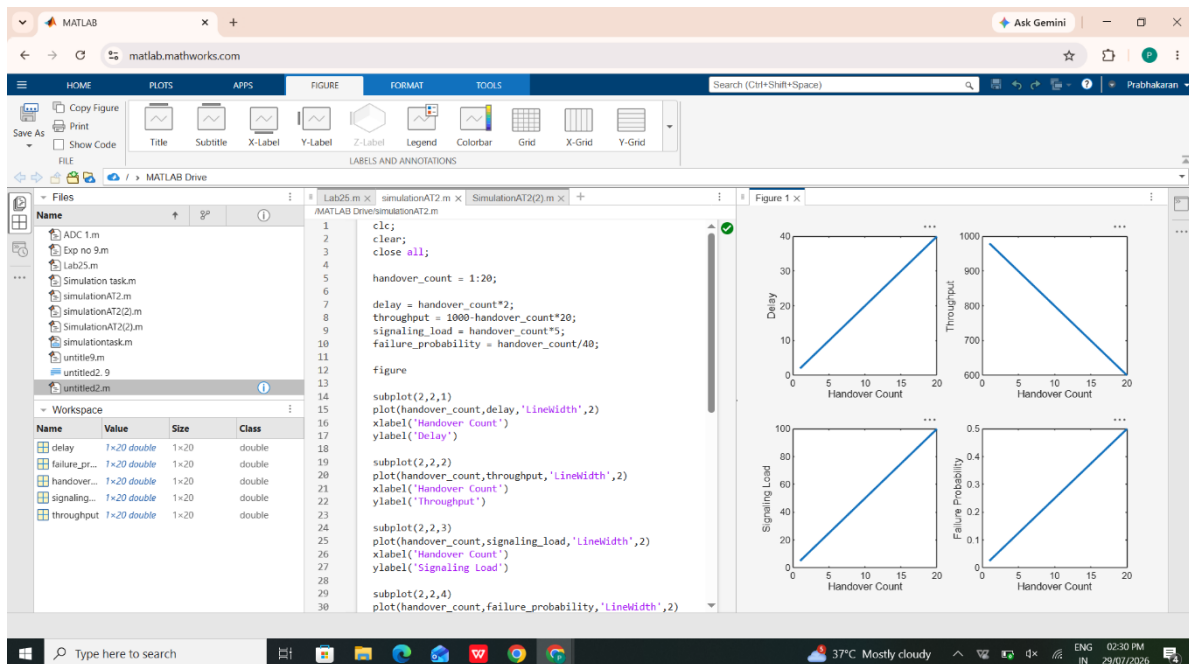
subplot(2,2,4)

plot(handover_count,failure_probability,'LineWidth',2)

xlabel('Handover Count')
```

ylabel('Failure Probability')

OUTPUT:



5. Simulation of Propagation Loss under 700 MHz, 3.5 GHz, 28 GHz, 38 GHz, and 60 GHz.

Aim: To simulate and compare propagation loss at different 5G operating frequencies (700 MHz, 3.5 GHz, 28 GHz, 38 GHz, and 60 GHz) using MATLAB.

Program:

clc;

clear;

close all;

distance = 1:1:1000;

f1 = 700;

f2 = 3500;

f3 = 28000;

f4 = 38000;

f5 = 60000;

PL700 = 32.4 + 20*log10(distance) + 20*log10(f1);


```

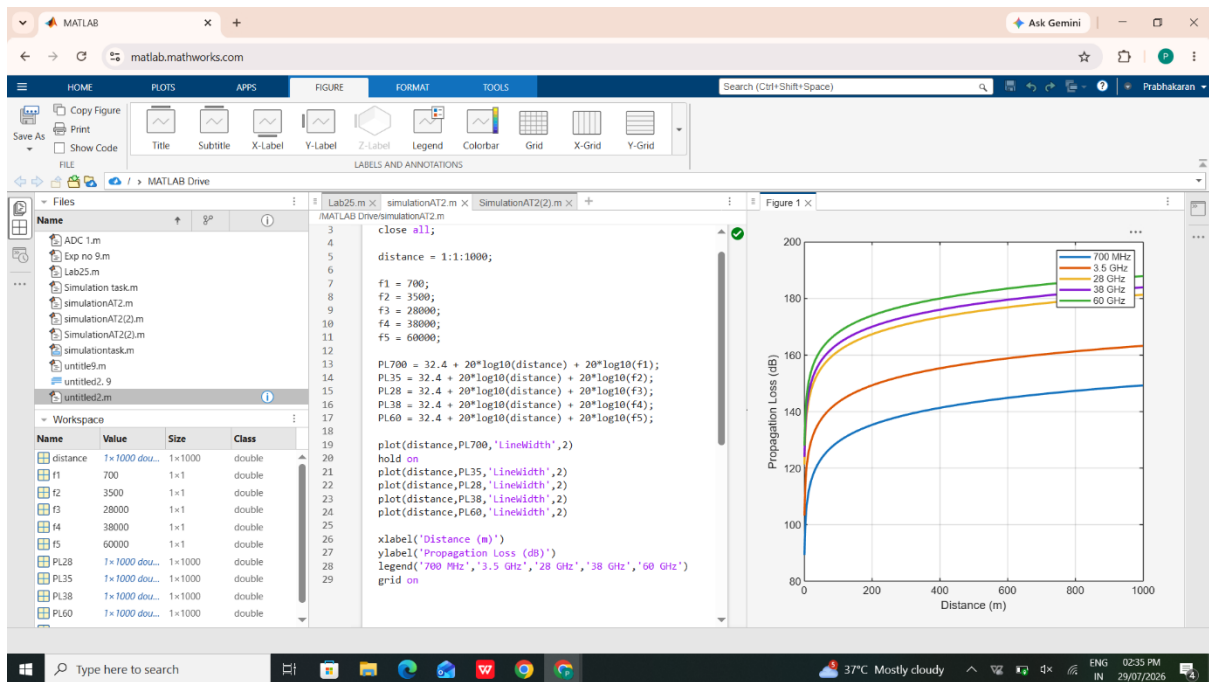
PL35 = 32.4 + 20*log10(distance) + 20*log10(f2);
PL28 = 32.4 + 20*log10(distance) + 20*log10(f3);
PL38 = 32.4 + 20*log10(distance) + 20*log10(f4);
PL60 = 32.4 + 20*log10(distance) + 20*log10(f5);

plot(distance,PL700,'LineWidth',2)
hold on
plot(distance,PL35,'LineWidth',2)
plot(distance,PL28,'LineWidth',2)
plot(distance,PL38,'LineWidth',2)
plot(distance,PL60,'LineWidth',2)

xlabel('Distance (m)')
ylabel('Propagation Loss (dB)')
legend('700 MHz','3.5 GHz','28 GHz','38 GHz','60 GHz')
grid on

```

OUTPUT:



RESULT:

The simulation was successfully completed in MATLAB. The results demonstrated that propagation loss increased with both frequency and transmission distance. Lower-frequency bands such as 700 MHz experienced the least path loss, whereas higher-frequency bands such as 28 GHz, 38 GHz, and 60 GHz exhibited significantly greater propagation loss.